Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/27/2025
PI Name:	Driskell, Tripp Ph.D.		
Project Title:	Crew Adaptive Decision Making Under Stress (CADMUS) and Crew Decision Support System (CDSS): Development, Validation and Proof-of-Concept		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (2) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32169	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 HERO 80JSC024NA001-FLAGSHIP: 2024 HERO Appendix A: NASA Human Research Program Flagship Opportunity
Start Date:	03/03/2025	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per A. Beitman/HFBP. Original end date was 03/02/2030 (Ed., 8/14/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Burke, Shawn Ph.D. (University Of Central Florida) Driskell, James Ph.D. (Florida Maxima Corp)		
Grant/Contract No.:	80NSSC25K7922		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Aims. The focus of the proposed effort is on: (1) enhancing the problem-solving abilities of the crew, as well as the extended team (Front Room, Back Room, and Mission Evaluation Room Controllers), during off-nominal conditions; and (2) developing a prototype Crew Decision Support System (CDSS), aimed at assisting decision making and problem solving activities given operational constraints (e.g., distributed teams, communication delay). The first focus, enhancing crew problem-solving abilities, will be achieved through the development and validation of the Crew Adaptive Decision Making Under Stress (CADMUS) training. The CADMUS training will adopt the validated approach of Stress Exposure Training (SET), which provides a framework for designing, developing, and implementing training for off-nominal or high-demand events. This comprehensive training program will incorporate a continuum of graduated training activities to support adaptive crew decision making under stress – that are geared toward Artemis-class missions. The CDSS will be a model decision support system designed to facilitate crew problem-solving and decision making skills, given the operational constraints of Artemis missions (e.g., distributed teamwork and varied communication delays). Methods. The Crew Adaptive Decision Making Under Stress (CADMUS) training unites multiple streams of research to enhance the problem-solving abilities of the crew, as well as the extended team (Front Room, Back Room, and Mission Evaluation Room Controllers), during off-nominal conditions. Specifically, CADMUS training will integrate the Team Macrocognition and Stress Exposure Training (SET) literatures. In brief, this proposed effort will identify the macrocognitive processes involved in Artemis-class mission collaborative decision making and problem-solving, with an eye towards the stressors present in this setting (e.g., comm delays, work pace) and the types of problems crews may confront (e.g., type, structure). The outcome will be a set of contextualized metacognitive crew processes that will serve as the input for a training intervention applying the SET structure. We will then utilize the Human Exploration Research Analog (HERA) campaigns 8 and 9 to test and validate this approach. Moreover, to bolster validation efforts, we will carry out laboratory research, as well as field research in an additional spaceflight analog (LunAres Research Station). Concurrently, we will develop a prototype Crew Decision Support System (CDSS) tool, aimed at assisting decision making and problem-solving activities given operational constraints (e.g., distributed teams, communication delay), and test this in the HERA analog. The contextualized metacognitive crew processes will also serve as input for the development of the CDSS. The resultant effort will establish the requirements for the building of a CDSS to facilitate in-mission problem-solving, mapped to the contextualized metacognitive crew processes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Contextualized countermeasures and tools aimed at preventing process issues during decision-making and promoting problem-solving capabilities will support mission success. This research will utilize HERA campaigns 8 and 9 to test and validate novel approaches targeted to lunar missions. The results will deliver new techniques and tools to reduce risks and associated gaps identified by the NASA Human Research Program.
Task Progress:	December 2025 update per the PI (Ed., 1/6/25) Antarctica participant data: 1000 journal entries, 400 bi-weekly surveys. Topics: work relations, friendship relations, close relationships, general subjects. --- Primary tasks in the current reporting period included examining prior research that has examined the types of decision and problem solving events that occur within spaceflight, especially those that might be expected or have been shown to occur on lunar missions. The goal of this task is to identify the types of events that might occur within Artemis class missions which require complex problem solving, as well as the corresponding macrocognitive processes that are involved in the enactment of problem solving in these contexts. Following, we began to categorize the nature of decision profiles within Artemis class missions; specifically, using the information gained to build decision profiles indicative of effective decision making in Artemis class missions. These profiles specify the types and expected levels [high, medium, low] of the macrocognitive processes involved in emergent problem sets and the degree to which the process is distributed [both within the crew and across the crew and mission control]. The outcome of these primary tasks was to develop a framework that categorizes the types of problem events and associated problem solving processes utilized most often as mapped to each Artemis phase (i.e., pre-, in-, and post-mission). A key focus of this reporting period was to finalize our research design, study materials, and measures in order to conduct research in NASA's Human Exploration Research Analog (HERA) Campaign 8. The study was to be conducted at NASA Johnson Space Center by NASA Human Research Program. The current study, utilizing HERA Campaign 8, is a control arm aimed at documenting the effects of perceived stress on individual and team functioning. The baseline data gathered in Campaign 8 would be used as a comparison for future data collection. The primary aim of this study was to develop and validate the Crew Adaptive Decision Making Under Stress (CADMUS) training. This research aimed to deliver stand-alone data, scientific knowledge, and analog validated countermeasures to narrow the identified NASA Human Research Roadmap (HRR) gaps: • Team Gap 105: We need to identify a set of countermeasures to support team function and enable multiple distributed teams to manage shifting levels of autonomy for all phases of increasingly earth independent, long duration exploration missions. • Team Gap 104: We need to identify validated ground-based and in-flight training methods for both preparatory and sustaining team function during shifting autonomy in increasingly Earth independent, long duration exploration missions. This research project was terminated at the beginning of the proposed research efforts and before data collection.
Bibliography Type:	Description: (Last Updated:)